Silvanius Brian Mulu

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Education

Texas State University San Marcos, TX
B.S. Applied Mathematics (Honors), B.A. Economics May 2029

- GPA: 4.0
- Relevant Coursework: Calculus I, II, III; Applied Linear Algebra; Probability & Statistics; (Honors) Intermediate MacroEconomics

Experience

Texas State University, Department of Economics & Finance San Marcos, TX
Undergraduate Research Assistant Jan 2026 – Present
Technologies: R, MATLAB, LaTeX [Code] | [Work-In-Progress]

- Estimating industry level transmission of oil price shocks using a **structural input output VAR**, decomposing spot and speculative components and computing **impulse response functions** for inflation, output and trade balance
- Applying **regression discontinuity** designs to SNAP eligibility thresholds to identify causal effects on household relocation

National Social Security Fund Nairobi, Kenya
Data Analyst Intern Mar 2025 – Jun 2025
Technologies: Power BI, Excel, Python

- Built **Power BI** dashboards to monitor payment flows and portfolio profitability across the national pension fund portfolio, enabling real time performance tracking
- Collaborated with backend engineers on **schema design** and **data validation** rules for the member registration system

Projects

Independent Study Thesis: Volatility Regimes in Cointegrated US Energy Pairs [Work-In-Progress]
Technologies: R, LaTeX (Work-In-Progress)

- Developing a two state **Markov switching model** with **time varying transition probabilities** to detect when cointegrated pairs across a CRSP energy universe (2000 - 2025) shift between calm and turbulent volatility regimes testing how turbulence spreads and whether it precedes relationship breakdowns

Statistical Arbitrage: Pairs Trading in U.S. Oil Equities [Code] | [Paper]
Technologies: R, Python, LaTeX (Work-In-Progress)

- Tested **cointegration based pairs trading** on US energy equities, showing that a widely cited pair fails under correctly sized **Engle Granger** critical values and that apparent risk adjusted outperformance comes from hedging driven variance reduction rather than mean reversion.

Leadership & Extracurriculars

Financial Management Association San Marcos, TX
Vice President, Quant Finance Cohort August 2026 – Present

- Lead quantitative markets education and technical workshops for club members while supporting recruitment and professional development.

International Student Organisation San Marcos, TX
Treasurer, Head of Finance Committee, Board of Directors Sept 2024 – Present

- Oversee annual budget and financial planning for a 1,500+ member organization, ensuring fiscal accountability and strategic fund allocation.

Awards

Shelby Davis Scholar: Full tuition, Texas State
President's List (2X): Perfect GPA
Goldman Sachs: Possibilities Series Participant (2026)
College Fed Challenge: 1/5 Selected Team Member (2026)
Beta Gamma Sigma: Invited Member, Honor Society
MIT Policy Hackathon: Globalist Finalist (2024)
Jump Trading Probability Cup: Top 5% forecaster (2026)

Skills & Interests

- **Research Interests:** Time Series Econometrics, Energy Economics, Empirical Macroeconomics
- **Programming & Tools:** R, Python, JavaScript, Excel, Power BI, Git, GitHub, Linux.
- **Languages:** English (Native), Swahili (Native), Kikamba (Native).
- **Interests:** Long-distance running, Table Tennis (Collegiate Bronze, 6x High School Gold), Poker, Chess, Golf, Farming (300 chickens).